

Moiré-pattern-modulated electronic structures in Sb₂Te₃/graphene heterostructure

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ABSTRACT

Moiré superlattice has recently been found in topological insulators, which can lead to periodic modulation on the electronic structure. In this work, we report the low-temperature scanning tunneling microscopy study of Sb₂Te₃ films grown on graphitized 4H-SiC. We find that substrate temperature can strongly influence the rotation angles between Sb₂Te₃ film and graphene substrate. Three kinds of moiré patterns are observed at the first quintuple layer Sb₂Te₃ film under different substrate temperatures. One shows complicated patterns with a rotation angle of nearly 0° relative to the substrate, another just exhibits simple 1 × 1 structure with a rotation angle of 30°. Other rotation angle like 8.2° is observed at higher substrate temperature as well, which is relatively rare. Comparison of the dI/dV curves from Sb₂Te₃ films with different moiré patterns indicates that the superstructure can offer degrees of freedom in tailoring electronic structure. This work may stimulate the further study on the moiré modulation to the electronic properties of topological insulators.

KEYWORDS

moiré pattern, topological insulator, electronic structure, scanning tunneling microscopy

1 Introduction

Moiré pattern results from the overlap of two lattices with a lattice mismatch or a small-angle twist [1]. Moiré system can induce exotic electronic properties such as correlated states in twisted bilayer graphene (TBG) [2–5], Hofstadter's butterfly pattern in graphene/hexagonal boron nitride (hBN) heterostructures, interlayer coupling induced charge redistribution [6] and spatially dependent band gap in transition metal dichalcogenides (TMD) [7]. In addition to graphene, hBN and TMD, another family of moiré superlattice can be found in topological insulators (TIs) [8, 9]. A TI have an insulating bulk gap and nontrivial gapless surface states because of strong spin-orbit coupling [10]. Recently, surface states of TIs have been predicted to be affected by moiré superlattices [11]. The topological nature of TI prevents gap from opening with time-reversal symmetry preserved, so that moiré surface states do not form minibands like in graphene and TMD. Instead, van Hove singularities (VHS) are expected to appear near the Fermi level and to be strongly correlated with high-temperature superconductivity. However, few experiments have reported moiré pattern influencing electronic properties of a TI.

Sb₂Te₃ is a heavily p-doped TI with its Fermi level buried in the bulk valence band [12]. Sb₂Te₃ has a quintuple layered structure in the [111] direction. The sequential five hexagonal

sheets Te-Sb-Te-Sb-Te are called a quintuple layer (QL). The Sb₂Te₃ films have been grown on different substrates, such as Si (111) [13, 14], SrTiO₃ (111) [15] and graphitized SiC [16]. Although there is a lattice mismatch between the Sb₂Te₃ and the substrates, experiments related to the possible moiré pattern are still lacking. Since the periodicity of moiré pattern is in the range from several nanometers to several tens of nanometers, scanning tunneling microscopy/spectroscopy (STM/STS) are powerful tools to study the moiré-related phenomenon at atomic scale.

In this work, we explore the moiré patterns on the Sb₂Te₃/graphene heterostructure by STM/STS. Depending on the substrate temperature, three kinds of patterns with different rotation angles are observed at the first QL of Sb₂Te₃ films, including 0°, 8.2° and 30°. The moiré pattern with a rotation angle of 8.2° is rarely observed. Therefore, the following discussion will be focused on the rotation angle of 0° and 30°. One of them exhibiting an obvious contrast variation originates from a moiré superlattice caused by lattice mismatch and charge transfer between Sb₂Te₃ and graphene, the other one displays as an ordinary 1 × 1 structure. The dI/dV curves in the two regions are very different. Our results suggest the moiré modulation is a feasible route to tune the electronic properties of TIs.

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2 Experimental

The experiments were performed in an ultrahigh vacuum (UHV) MBE-STM joint system (Unisoku-1300) with a base pressure of 3×10^{-10} Torr. The bilayer graphene was prepared on a 4H-SiC (0001) substrate by annealing at 900 °C for 10 min and then post-annealing at 1,250 °C for 10 min [17]. High purity Sb and Te were thermally co-evaporated from standard Knudsen cells at $T_{\text{Sb}} = 390$ °C and $T_{\text{Te}} = 225$ °C. The substrate temperature was kept at 160 °C, 190 °C, and 210 °C during growth. Five minutes of post annealing was conducted at the same temperature in the Te-rich atmosphere for a better-quality film [18]. The flux ratio of Te to Sb sources is around 10:1, and the typical growth rate of the Sb_2Te_3 thin films was about 0.25 QL/min. Prepared samples were transferred to a cryogenic stage kept at 77 K for STM measurement. Electrochemically etched tungsten tips were used after flashing and silver decoration *in situ*. The dI/dV curves were taken via lock-in technique with a modulation of 5 mV at a frequency of 961 Hz. The STM images and fast Fourier transform (FFT) images processing was done with WSxM software.

3 Results and discussion

Figure 1(a) is a large-scale topographic STM image of Sb_2Te_3 films on graphitized 4H-SiC. The hexagonal atomic lattice of the graphene is displayed in Fig. 1(b). The lattice constant of graphene deduced from the FFT image is 0.246 nm. Figure 1(c) shows the line profiles cutting along the dark dash line drawn in Fig. 1(a), from which we can see that the terraces are atomically flat. The step heights are 1.2 nm for 1 QL and approximately 1.01 nm for other QLs, which is consistent with the previous reports. The dI/dV curves correspond to the local density of states for the film within the Tersoff-Hamann approximation [19]. Figure 1(d) shows the dI/dV curves measured on Sb_2Te_3 film with different thickness. The arrows marked in Fig. 1(d) correspond to the valence band maximum (VBM) and conduction band minimum (CBM) of the Sb_2Te_3 films, which agrees well with the reported values [12]. Moreover, bulk valence band states are M-shaped while bulk

conduction band states are nearly parabolic. The energy gap decreases with increasing film thickness, from $\sim 1,040$ meV at 1 QL to ~ 630 meV at 4 QL, which is much larger than the bulk band gap of 300 meV [16].

Recent experiments have shown the possibility of tuning the rotation angle by substrate temperature. For instance, selecting a substrate temperature between 1,530 and 1,000 K can lead to distinctive rotation angles 0° , $\pm 0.6^\circ$, $\pm 1.1^\circ$, and $\pm 1.7^\circ$ in the moiré pattern of graphene/Ir (111) heterostructure [20]. In our experiments, we find that rotation angles can also be strongly influenced by changing the substrate temperature. Sb_2Te_3 films grown on graphene have been shown in Figs. 2(a) and 2(b) at different substrate temperatures of 190 °C and 160 °C, respectively. A pronounced periodic structure containing bright dots and flower-like patterns can be clearly resolved in Fig. 2(c). The moiré pattern is distinct from the reported hexagonal structure of Sb_2Te_3 [12]. The rotation angle between Sb_2Te_3 and graphene is nearly 0° . The periodicity of the moiré pattern in Fig. 2(c) is 2.94 nm, which is calculated from the atomically resolved image in real space. The superstructure has been revealed by the FFT image in Fig. 2(e), in which the replicas of the inner hexagon are around the six bright Bragg points of the Sb_2Te_3 lattice. Faint signals can be discerned between the bright points. Signals from 12×12 of graphene are labeled in orange circles in the FFT images. The lattice constant of Sb_2Te_3 is 0.421 nm calculated from the FFT image. The moiré pattern is due to the lattice mismatch of 42.3% between graphene and Sb_2Te_3 . Generally, commensurate moiré patterns can be described as linear integer combinations of the lattice vectors of both the substrate and overlayer. The supercell is labeled as a rhombus consisted of white and black triangles and noted as $R(7, 12)$ [21]. The notation $R(7, 12)$ denotes that the overlayer Sb_2Te_3 unit cell length of 0.426 nm multiplied by 7, and the substrate graphene unit cell length of 0.246 nm multiplied by 12.

Figure 2(d) illustrates the atomically resolved image of 1 QL Sb_2Te_3 fabricated at the substrate temperature of 160 °C. Only 1×1 structure can be observed in the atomically resolved image and the corresponding FFT image in Fig. 2(f). The periodicity of 1 QL Sb_2Te_3 prepared at 160 °C is 0.426 nm. The twist angle is 30° inferred from the orientation of graphene substrate and Sb_2Te_3 films. As the lattice constant of Sb_2Te_3

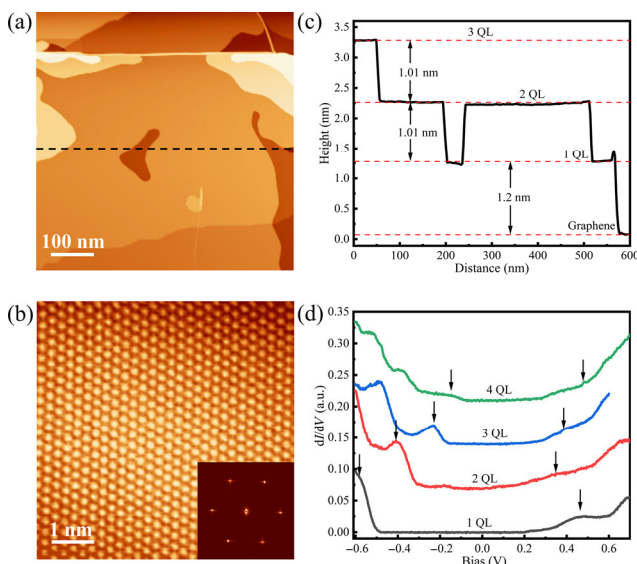


Figure 1 Topography and dI/dV curves of Sb_2Te_3 films grown on graphitized 4H-SiC (0001) substrate. (a) Topographic image of Sb_2Te_3 films grown on graphene ($V_b = 2.89$ V, $I = 100$ pA). (b) Atomically resolved images of graphene substrate ($V_b = 245$ mV, $I = 100$ pA). (c) Line profiles cutting along the black dash line marked in (a). (d) Series of dI/dV spectra measured on the Sb_2Te_3 films. Arrows indicate the energies of VBM and CBM of Sb_2Te_3 films.

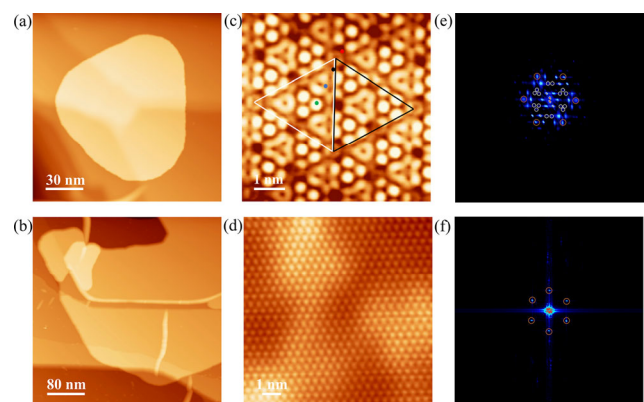


Figure 2 Atomically resolved STM images of Sb_2Te_3 films and graphene substrate. (a) and (b) Topographies of the Sb_2Te_3 films grown on the graphene substrate at 190 °C ($V_b = 2.79$ V, $I = 100$ pA) and 160 °C ($V_b = 2.18$ V, $I = 100$ pA), respectively. (c) and (d) Atomically resolved image of the moiré superstructures with rotation angles of 0° ($V_b = 86$ mV, $I = 100$ pA) and 30° ($V_b = -500$ mV, $I = 100$ pA). (e) and (f) The corresponding two-dimensional (2D) FFT images of (c) and (d). The 1×1 structure of Sb_2Te_3 is labeled by orange circles while the 12×12 structure of graphene is labeled by white circles.

(0.426 nm) is ~ 3 times of that of graphene (0.246 nm), the lattice points of Sb_2Te_3 will locate just on the hollow site of the honeycomb lattice of graphene. No contrast variation can be found since the atoms of Sb_2Te_3 are still equivalent. In fact, the rotation angles of the moiré modulations we observed in experiments not only include 30° at 160°C and 0° at 190°C , but also 8.2° at 210°C . The moiré modulation with rotation angles of 30° and 0° are easily found at corresponding substrate temperatures. Although the moiré modulation with rotation angle of 8.2° can be observed at substrate temperature of 210°C , the Sb_2Te_3 films tend to appear as 2 QL at this high substrate temperature and it is hard to find 1 QL Sb_2Te_3 film. Considering that the 8.2° moiré modulations are relatively rare, we only focus on the moiré modulations with rotation angles of 30° and 0° in the main text and show a representative example of 8.2° moiré modulation in the supplementary material (Fig. S1 in the Electronic Supplementary Material (ESM)). Moreover, moiré patterns with different rotation angles can coexist in our samples. The ratio of the observed moiré patterns with different rotation angles is influenced by substrate temperature. In experiments, we found that the ratio of the moiré patterns with a rotation angle of 0° to all the angles observed is changing from 1/13 to 2/3 when substrate temperature changed from 160 to 190°C . Besides, there is no moiré pattern observed in the atomically resolved STM images of 2 QL and the 3 QL Sb_2Te_3 . Thus, the modulation from the moiré pattern formed by Sb_2Te_3 and graphene should only survive at 1 QL Sb_2Te_3 on graphene.

In Fig. 3(a), the schematic illustration of Sb_2Te_3 surface atoms superimposed on that of graphene surface with a rotation angle of 30° is shown [22]. Atoms of Sb_2Te_3 lattice are represented by purple balls while atoms of graphene lattice are represented in orange balls. Atoms of Sb_2Te_3 lattice locate right above the hollow site of honeycomb lattice of graphene substrate. No contrast difference is visible in Fig. 3(a), which agrees well with Fig. 2(d). Figure 3(b) gives the schematic illustration produced according to the orientation of Sb_2Te_3 and graphene from the Figs. 2(c) and 1(b). The rotation angle is nearly 0° . The white and black triangles have been drawn to

represent the unit cell of the moiré superlattice. As we know, there is also another possible explanation for the contrast variation in the STM images, which is charge density wave (CDW). Hexagonal deformed Fermi surface and strong nesting found in Bi_2Te_3 and Bi_2Se_3 have led to several predictions of CDW in the TI heterostructure. However, no signature of CDW gap near the Fermi level was detected in the experiments, even at a low temperature of 4.2 K . The CDW origin of the long periodicity structure can be excluded in our experiments. In addition, there is no report about the CDW phenomena in Sb_2Te_3 .

Figure 3(c) shows two typical dI/dV curves measured at 1 QL Sb_2Te_3 with different orientations with respect to the substrate. The similar kinks within the band gap of 1 QL Sb_2Te_3 can be resolved in the two spectra, which are marked by the black arrows. The dI/dV spectra taken on the area with the rotation angle of 0° shows an energy shift of about 0.11 eV towards the Fermi level with respect to that of 30° . Generally, the lattice mismatch between Sb_2Te_3 and graphene can be defined by the simple formula:

$$f = \frac{a_s - a_e}{a_e} \quad (1)$$

where a_s and a_e are the lattice constant of the substrate and the overlayer, respectively [21]. The mismatch can take on either sign. Differences can be observed between tensile ($f > 0$) and compressive ($f < 0$) systems. The lattice mismatch between Sb_2Te_3 and graphene is 42.3% . However, the mismatch can be reduced by different stacking of the Sb_2Te_3 and graphene. For the 30° rotation case, no moiré pattern is observed and the mismatch

$$f = \frac{\sqrt{3} \times 0.246 - 1 \times 0.426}{1 \times 0.426} = 0.02\% \quad (2)$$

For the 0° case,

$$f = \frac{12 \times 0.246 - 7 \times 0.426}{7 \times 0.426} = 1\% \quad (3)$$

In our experiments, the observed rotations are associated with the favorable low-strain structures thus requiring higher temperature for the moiré pattern with almost no rotation. The difference of dI/dV curves between two epitaxial orientations is mainly ascribed to a combined effect of lattice mismatch difference and charge transfer. Similar phenomenon was observed in the system of $\text{MoS}_2/\text{graphite}$ [21]. Moiré pattern of three rotation angles 1° (R(37/61)), 14° (R(19/31)) and 20° (R(4/7)) were observed. R(37/61) and R(19/31) with close lattice mismatch share similar dI/dV curves with a shift on the peak positions. It is proposed by Huang et al. that the lattice strain can shift the band toward the Fermi level [23], which is consistent with the phenomenon observed in Fig. 3(c). On the other hand, charge transfer mechanism may also play an important role in the process [24]. When two materials come into contact, electrons will be transferred from one to the other until their Fermi levels are equal [25]. The work functions of graphene and Sb_2Te_3 are 4.5 eV [26] and 5.33 eV [27]. Electrons will be transferred from graphene to Sb_2Te_3 , and the band structure will be shifted towards negative direction. As charge transfer decays exponentially with distance [28], the effects of substrate doping differ in different epitaxial directions. The distance between the substrate and the valley sites of the overlayer is smaller than that of the peak sites, so that the charge transfer is stronger in the valley sites than in the peak sites. When the rotation angles is 30° , the Te atoms tend to locate at the hollow sites of honeycomb for a lower

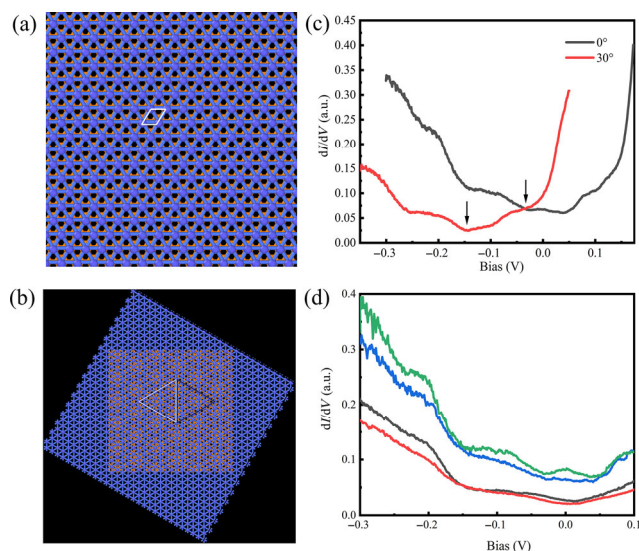


Figure 3 Schematic illustration and electronic structure of the $\text{Sb}_2\text{Te}_3/\text{graphene}$ heterostructure. (a) and (b) Schematic illustration of Sb_2Te_3 surface atoms superimposed on that of graphene surface with a rotation angle of 30° and 0° , respectively. The unit cells have been labeled in white and black. (c) dI/dV curves taken at the areas with rotation angle of 0° (setpoint: $I = 0.2\text{ nA}$) and 30° (setpoint: $I = 0.1\text{ nA}$), respectively. (d) dI/dV curves measured on the location marked with the green, blue, black, and red dots in Fig. 2(c) (setpoint: $I = 0.2\text{ nA}$).

energy. In this case, the charge transfer is the strongest. The height of atoms at different sites in the moiré pattern between Sb_2Te_3 and graphene with no rotation are different. Once the Te atoms move to other locations rather than hollow sites, charge transfer will be weakened. Therefore, the dI/dV curve of 30 shifts towards negative direction with respect to that of 0° .

We are also interested in how the electronic structure varies within a moiré superlattice cell. We measure the dI/dV curves of the bright and dark areas marked in green, blue, red, and black shown in Fig. 2(c) at a tunneling current of 0.2 nA. In Fig. 3(d), we observe that the kinks obtained in the bright areas have a larger signal than those in the dark areas. As we mentioned before, charge transfer decays exponentially with distance. The dI/dV curves are affected by the corresponding stacking of the Te atoms with respect to the C atoms. More electrons will be transferred from the substrate to valley sites than peak sites, thus presenting the contrasts in Fig. 2(c). It has been predicted that high order VHS will appear in moiré superlattice in topological insulators [11]. However, in our experiments we did not observe such a high order VHS in tunneling spectra measured at the 1 QL Sb_2Te_3 /graphene. One possible explanation is that the nearly flat-band induced high order VHS cannot be formed at 1 QL Sb_2Te_3 /graphene moiré superlattice, for which the flat-band are resulting from the moiré modulated topological surface states. The topological surface states of Sb_2Te_3 start to form at a thickness of 4 QL, and there are no topological surface states at 1 QL Sb_2Te_3 /graphene heterostructure. The absence of VHS may be related to the absence of surface states at 1 QL. Another explanation is that the VHS may only exist at moiré superlattice with certain rotation angles, which are not match with these superlattices we observed in experiments.

Although moiré pattern at the surface disappears when the thickness of Sb_2Te_3 is above 2 QL, modulation from moiré pattern may still exist in the interface. Because the rotation angle between 1 QL Sb_2Te_3 and graphene will not change with the increase of thickness. However, STM measurements are limited to a few atomic layers at the surface, where the interfacial effects are hardly probed. Recently, moiré pattern with periodicity of ~ 8.9 nm has been observed by the transmission electron microscopy (TEM) at the graphene/Al (111) interface [29]. A non-atomic lattice formed by interlayer coupling induced charge redistribution in Bi_2Se_3 /TMD heterostructures has been revealed by high-energy selected area electron diffraction (SAED) as well [6]. Our experiment is a very first attempt to investigate the moiré modulation effects to the electronic properties of topological insulators. More detailed investigation on the moiré superlattice at interface, including transport experiments and other characterization methods is required.

4 Conclusion

To summarize, we have explored the structural and electronic properties of Sb_2Te_3 /graphene heterostructures. We find that rotation angles between TI and graphene can be strongly influenced by changing substrate temperature. Moiré pattern of the Sb_2Te_3 /graphene heterostructures with distinct rotation angles brings energy shift in the dI/dV curves. In addition, the electronic structures difference of the bright and dark areas at 1 QL Sb_2Te_3 may come from lattice mismatch difference and charge transfer from substrate. The modulation from moiré pattern can only survive at 1 QL so that it has no impact on the surface states of Sb_2Te_3 . As suggested by Wang et al. [11], superconductivity may occur on moiré surface states of TIs with transition temperature up to $T_c \sim 10$ K. Further research on electronic properties of the moiré pattern at the interface is

suggested. Our project on the moiré pattern consisting of TIs and two-dimensional materials provide a platform to detect the topological superconductivity.

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References

- [1] Pong, W. T.; Durkan, C. A review and outlook for an anomaly of scanning tunnelling microscopy (STM): Superlattices on graphite. *J. Phys. D Appl. Phys.* **2005**, *38*, R329–R355.
- [2] Balents, L.; Dean, C. R.; Efetov, D. K.; Young, A. F. Superconductivity and strong correlations in moiré flat bands. *Nat. Phys.* **2020**, *16*, 725–733.
- [3] Andrei, E. Y.; MacDonald, A. H. Graphene bilayers with a twist. *Nat. Mater.* **2020**, *19*, 1265–1275.
- [4] Cao, Y.; Fatemi, V.; Fang, S. A.; Watanabe, K.; Taniguchi, T.; Kaxiras, E.; Jarillo-Herrero, P. Unconventional superconductivity in magic-angle graphene superlattices. *Nature* **2018**, *556*, 43–50.
- [5] Carr, S.; Fang, S. A.; Kaxiras, E. Electronic-structure methods for twisted moiré layers. *Nat. Rev. Mater.* **2020**, *5*, 748–763.
- [6] Hennighausen, Z.; Lane, C.; Buda, I. G.; Mathur, V. K.; Bansil, A.; Kar, S. Evidence of a purely electronic two-dimensional lattice at the interface of TMD/ Bi_2Se_3 heterostructures. *Nanoscale* **2019**, *11*, 15929–15938.
- [7] Zhang, C. D.; Chuu, C. P.; Ren, X. B.; Li, M. Y.; Li, L. J.; Jin, C. H.; Chou, M. Y.; Shih, C. K. Interlayer couplings, moiré patterns, and 2D electronic superlattices in $\text{MoS}_2/\text{WSe}_2$ hetero-bilayers. *Sci. Adv.* **2017**, *3*, e1601459.
- [8] Song, C. L.; Wang, Y. L.; Jiang, Y. P.; Zhang, Y.; Chang, C. Z.; Wang, L. L.; He, K.; Chen, X.; Jia, J. F.; Wang, Y. Y. et al. Topological insulator Bi_2Se_3 thin films grown on double-layer graphene by molecular beam epitaxy. *Appl. Phys. Lett.* **2010**, *97*, 143118.
- [9] Schouteden, K.; Li, Z.; Chen, T. S.; Song, F. Q.; Partoens, B.; Van Haesendonck, C.; Park, K. Moiré superlattices at the topological insulator Bi_2Te_3 . *Sci. Rep.* **2016**, *6*, 20278.
- [10] Hasan, M. Z.; Kane, C. L. *Colloquium: Topological insulators. Rev. Mod. Phys.* **2010**, *82*, 3045–3067.
- [11] Wang, T.; Yuan, N. F. Q.; Fu, L. Moiré surface states and enhanced superconductivity in topological insulators. *Phys. Rev. X* **2021**, *11*, 021024.
- [12] Jiang, Y. P.; Sun, Y. Y.; Chen, M.; Wang, Y. L.; Li, Z.; Song, C. L.; He, K.; Wang, L. L.; Chen, X.; Xue, Q. K. et al. Fermi-level tuning of epitaxial Sb_2Te_3 thin films on graphene by regulating intrinsic defects and substrate transfer doping. *Phys. Rev. Lett.* **2012**, *108*, 066809.
- [13] Plucinski, L. P.; Herdt, A.; Fahrendorf, S.; Bihlmayer, G.; Mussler, G.; Döring, S.; Kampmeier, J.; Matthes, F.; Bürgler, D. E.; Grützmacher, D. et al. Electronic structure, surface morphology, and topologically protected surface states of Sb_2Te_3 thin films grown on Si(111). *J. Appl. Phys.* **2013**, *113*, 053706.
- [14] Wang, G.; Zhu, X. G.; Wen, J.; Chen, X.; He, K.; Wang, L. L.; Ma, X. C.; Liu, Y.; Dai, X.; Fang, Z. et al. Atomically smooth ultrathin films of topological insulator Sb_2Te_3 . *Nano Res.* **2010**, *3*, 874–880.

- [15] Zhang, T.; Ha, J.; Levy, N.; Kuk, Y.; Strosio, J. Electric-field tuning of the surface band structure of topological insulator Sb_2Te_3 thin films. *Phys. Rev. Lett.* **2013**, *111*, 056803.
- [16] Jiang, Y. P.; Wang, Y. L.; Chen, M.; Li, Z.; Song, C. L.; He, K.; Wang, L. L.; Chen, X.; Ma, X. C.; Xue, Q. K. Landau quantization and the thickness limit of topological insulator thin films of Sb_2Te_3 . *Phys. Rev. Lett.* **2012**, *108*, 016401.
- [17] Hass, J.; De Heer, W. A.; Conrad, E. H. The growth and morphology of epitaxial multilayer graphene. *J. Phys. Condens. Matter* **2008**, *20*, 323202.
- [18] Xu, J. P.; Liu, C. H.; Wang, M. X.; Ge, J. F.; Liu, Z. L.; Yang, X. J.; Chen, Y.; Liu, Y.; Xu, Z. A.; Gao, C. L. et al. Artificial topological superconductor by the proximity effect. *Phys. Rev. Lett.* **2014**, *112*, 217001.
- [19] Tersoff, J.; Hamann, D. R. Theory of the scanning tunneling microscope. *Phys. Rev. B Condens Matter* **1985**, *31*, 805–813.
- [20] Omambac, K. M.; Hattab, H.; Brand, C.; Jnawali, G.; N'Diaye, A. T.; Coraux, J.; Van Gastel, R.; Poelsema, B.; Michely, T.; Heringdorf, F. J. M. Z. et al. Temperature-controlled rotational epitaxy of graphene. *Nano Lett.* **2019**, *19*, 4594–4600.
- [21] Lu, C. I.; Butler, C. J.; Huang, J. K.; Chu, Y. H.; Yang, H. H.; Wei, C. M.; Li, L. J.; Lin, M. T. Moiré-related in-gap states in a twisted MoS_2 /graphite heterojunction. *npj 2D Mater. Appl.* **2017**, *1*, 24.
- [22] Hermann, K. Periodic overlayers and moiré patterns: Theoretical studies of geometric properties. *J. Phys. Condens. Matter* **2012**, *24*, 314210.
- [23] Huang, Y. L.; Chen, Y. F.; Zhang, W. J.; Quek, S. Y.; Chen, C. H.; Li, L. J.; Hsu, W. T.; Chang, W. H.; Zheng, Y. J.; Chen, W. et al. Bandgap tunability at single-layer molybdenum disulphide grain boundaries. *Nat. Commun.* **2015**, *6*, 6298.
- [24] Shao, Z. B.; Fu, Z. G.; Li, S. J.; Cao, Y.; Bian, Q.; Sun, H. G.; Zhang, Z. Y.; Gedeon, H.; Zhang, X.; Liu, L. J. et al. Strongly compressed few-layered SnSe_2 films grown on a SrTiO_3 substrate: The coexistence of charge ordering and enhanced interfacial superconductivity. *Nano Lett.* **2019**, *19*, 5304–5312.
- [25] Zhang, Y. M.; Fan, J. Q.; Wang, W. L.; Zhang, D.; Wang, L. L.; Li, W.; He, K.; Song, C. L.; Ma, X. C.; Xue, Q. K. Observation of interface superconductivity in a SnSe_2 /epitaxial graphene van der Waals heterostructure. *Phys. Rev. B* **2018**, *98*, 220508.
- [26] Giovannetti, G.; Khomyakov, P. A.; Brocks, G.; Karpan, V. M.; Van Den Brink, J.; Kelly, P. J. Doping graphene with metal contacts. *Phys. Rev. Lett.* **2008**, *101*, 026803.
- [27] Kim, S.; Mo, J. H.; Jang, K. S. High-performance n-type carbon nanotubes doped by oxidation of neighboring Sb_2Te_3 for a flexible thermoelectric generator. *ACS Appl. Mater. Interfaces* **2020**, *12*, 43778–43784.
- [28] Wang, Y. L.; Jiang, Y. P.; Chen, M.; Li, Z.; Song, C. L.; Wang, L. L.; He, K.; Chen, X.; Ma, X. C.; Xue, Q. K. Scanning tunneling microscopy of interface properties of Bi_2Se_3 on FeSe . *J. Phys. Condens. Matter* **2012**, *24*, 475604.
- [29] Zhang, S.; He, D. J.; Huang, P.; Wang, F. Moiré pattern at graphene/ Al (111) interface: Experiment and simulation. *Mater. Des.* **2021**, *201*, 109509.